

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EE)

April/May 2012

EE 208 Micro Electronics & Integrated Circuit

Date: 04.05.2012, Friday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Explain summing, scaling and averaging amplifier in inverting configuration. [07]

Q - 2 (a) Define the following terms: [05]

- (i) Common mode rejection ratio
- (ii) Power supply rejection ratio
- (iii) Slew rate
- (iv) Bandwidth
- (v) Output resistance

(b) Explain working of IC 555 as an astable multivibrator. [05]

(c) Explain block diagram of op-amp. [04]

OR

Q - 2 (a) Explain Triangular wave generator. [06]

(b) Explain pin configuration of 741 op-amp IC. [04]

(c) Explain integrator using op-amp. [04]

Q - 3 (a) Define the following: [03]

- (i) Pass band
- (ii) Stop band
- (iii) Cut off frequency

(b) Explain differential Amplifier with one op-amp. [04]

(c) Explain features of op-amp & describe closed loop gain for non-inverting. [07]

OR

Q - 3 (a) Draw complete pin-diagram of 555 timer IC and explain working of each pin. [06]

(b) Define filter and Write merits and demerits of Active filters over passive filters. [04]

(c) Draw and explain Phase shift oscillator. [04]

SECTION – II

- Q - 4 Explain Common Emitter connection of a Transistor. Derive expression for β (in terms of α) and Collector current I_C . Explain concept of I_{CEO} with neat diagram. [07]
- Q - 5 (a) For the common base circuit shown in **Figure 1**, determine I_C and V_{CB} . Assume the transistor to be of silicon. [04]
- (b) Explain biasing with Feedback Resistor method with neat diagram. Also write down its Advantages and Disadvantages. [06]
- (c) What are the differences between JFET and BJT? [04]

OR

- Q - 5 (a) Find the h parameters of the circuit shown in **Figure 2**. [04]
- (b) Explain Potential Divider biasing of JFET. [04]
- (c) Describe h parameter of a transistor and its circuit performance. [06]
- Q - 6 (a) **Figure 3** shows the voltage divider bias method. Draw the D.C. load line and determine the operating point. Assume transistor to be of silicon. [07]
- (b) Define the following terms for JFET with necessary circuit and graph. [07]
- (i) Shorted-gate drain current (I_{DSS})
- (ii) Pinch off voltage (V_P)
- (iii) Gate-source cut off voltage ($V_{GS(off)}$)

OR

- Q - 6 (a) Define following parameters of JFET. [06]
- (i) a.c. drain resistance (r_d)
- (ii) Transconductance (g_m)
- (iii) Amplification factor (μ)
- (b) Write down advantages of JFET. [04]
- (c) Draw the h parameter equivalent circuit and Determine h parameter equation. [04]


